

Knapsack

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Solution of the practice quiz

We introduce a binary variable x_i for each item. It is one if the item is included in the sack, and 0 otherwise.

The knapsack problem is formulated as follows:

$$\max_{x \in \{0,1\}^4} \sum_{i=1}^4 u_i x_i,$$

subject to:

$$\sum_{i=1}^4 w_i x_i \leq W,$$
$$x_i \in \{0,1\}, \quad i = 1, \dots, 4,$$

where u_i is the utility of item i , w_i is the weight of item i and W is the capacity of the sack. In our case,

$$\max_{x \in \{0,1\}^4} x_1 + 4x_2 + 5x_3 + 7x_4,$$

subject to

$$x_1 + 3x_2 + 4x_3 + 5x_4 \leq 7,$$
$$x_i \in \{0,1\}, \quad i = 1, \dots, 4.$$

To identify the best sack, we enumerate all possible solutions. There are $2^4 = 16$ of them. For each of them, we calculate their total weight, and total utility. Then, we select the combination associated with the highest total utility among those that verify the capacity constraint.

x_1	x_2	x_3	x_4	$u^T x$	$w^T x$	Feasible?
0	0	0	0	0	0	YES
0	0	0	1	7	5	YES
0	0	1	0	5	4	YES
0	0	1	1	12	9	NO
0	1	0	0	4	3	YES
0	1	0	1	11	8	NO
0	1	1	0	9	7	YES
0	1	1	1	16	12	NO
1	0	0	0	1	1	YES
1	0	0	1	8	6	YES
1	0	1	0	6	5	YES
1	0	1	1	13	10	NO
1	1	0	0	5	4	YES
1	1	0	1	12	9	NO
1	1	1	0	10	8	NO
1	1	1	1	17	13	NO

The optimal solution consists in taking water and food, and no laptop and no clothes. The total utility is 9, and the total weight is 7.